

MGT 403: Business Analytics for Managers *Fall 2025*

PROFESSOR: Sanjiv Erat

EMAIL: serat@ucsd.edu

PHONE: (858) 822-0314

OFFICE HOURS: by appt through zoom.

Note: This syllabus may be updated and revised at a later date.

COURSE DESCRIPTION

In today's increasingly digital economy, the availability of data is widespread and business success can depend on smart, capable, and clever use of this data. Business analytics now drives ticket pricing, retail assortments, movie recommendations, digital marketing campaigns, healthcare scheduling, crime fighting, social media investments, capacity decisions and more. Today's companies generate an immense volume of transactional data, capturing trillions of bytes of information about their customers, suppliers, and operations. Millions of networked sensors are also embedded in the physical world in devices such as computers, mobile phones, smart energy meters, automobiles and industrial machines. Further, in a digitized world, consumers communicate, browse, buy, share, and search, creating extensive trails of valuable data.

Hence, analytics — the use of data with statistical and quantitative models to make better, data-driven business decisions — has become important to gain competitive advantages. With all of these different sources of data ranging from online sensors, email campaigns, to physical sensors, business analytics is pervasive in any size business.

This importance of analytics means that all managers should be comfortable discussing, assessing, and effectively conducting their own analyses. There is a significant shortage (and significant demand!) of business analysts and project managers who are adept at effectively utilizing analytics to create substantial value to an organization by solving impactful business problems.

This course introduces students to the foundations of business analytics. It teaches managers how to be comfortable exploring their data, how to create and use models to make better decisions in complex settings, and how to use statistics to better understand their business setting. Prior analytical experience is not a prerequisite for this course, we

will introduce all of the topics throughout this course, and students of all different backgrounds have found the class useful and approachable.

LEARNING OBJECTIVES

There are three main Units we cover in this course. Each Unit teaches you tools to answer key data- driven and quantitative questions. Each of the three units center around a central theme of addressing and studying uncertainty. Data and business analytics allows us to better understand and deal with uncertainty.

Unit 1: Randomness and Uncertainty

- Decisions when facing uncertain and probabilistic (“random”) future outcomes.
- Test the important of our assumptions in our models (“sensitivity analysis”)
- Modeling randomness to understand what risks and potential outcomes you face.
- Understanding why data gives us confidence and interpreting such confidence.
- Making evidence-based business decisions (“hypothesis testing”).
- Is the evidence sufficient to convince us? Or is it just noise? [Key business question.]

Unit 2: Prediction and modeling relationships

Unit 1 primarily focuses on studying individual random variables. Unit 2 expands our tools to model relationships between multiple variables.

- Use data to create predictions for the future (“regression”).
- Use regression to interpret the relationships between data.
 - o Understand the limitations of regression and how to address those limitations.
 - o Understand the limitations of causality interpretation.
- Use simulation to predict complex future outcomes (such as long-tail insurance risks).

Unit 3: Optimization

In optimization, we transition from modeling random outcomes to optimizing decisions, such as how many resources to allocate to different sectors or investments. Each use of resources has tradeoffs, which we learn to use tools to quantitatively model and consider.

- Simulating possible futures, and using this simulation to make the optimal decision.
- Formalizing the resource tradeoffs you face in a business into an optimization model.
- Using optimization model to understand the best use of many scarce resources, when faced with many tradeoffs.

All of these are KEY ways businesses and managers must think about their problems. This course will give you ample experience in these fundamental ways of quantitative thinking.

RADIANT TOOL

The central tool we use in this course to perform analysis is Radiant.

The modern use of statistics and analytics heavily relies on statistical software to conduct analyses. “R” is one of the most famous and is heavily used throughout all companies. However, using R requires being comfortable with coding first.

To best teach analytical thinking, we use Radiant to give you the power of exploring your data, visualizing it, and seeing analysis without needing to code. Radiant is built on-top of R, meaning it is free and it can be used at any business that can use R — which again, almost all modern analytics teams support the use of R.

Lastly and most importantly, Radiant is a valuable resume piece. This course will not only give you comfort with “data visualization”, “business analytics”, “regression analysis”, and “linear optimization” — you will also have experience with “R statistical analysis (via the Radiant interface)”.

MATERIALS

Required textbook (provided digitally, though physical versions can be ordered online)

- Data, Models, and Decisions: The Fundamentals of Management Science, by Dimitris Bertsimas and Robert M. Freund, 2004, Dynamic Ideas, Belmont, MA, ISBN 0-9759146-0-X.

SCHEDULE

Date	Class Topic & Activities
Week 1 (Unit 1)	Decision Analysis, Sensitivity Testing
Week 2	Modeling Probability Distributions, Conditional Probability
Week 3	Hypothesis Testing, Confidence Intervals
Week 4 (Unit 2)	Introduction to Regression Predictive Analytics
Week 5	Regression Tricks and Trips
Week 6	Using Simulation to Analyze Complex Probabilities
Week 7	— <i>Midterm Presentations</i> —
Week 8 (Unit 3)	Advanced Simulation – Optimizing Decisions
Week 9	Introduction to Linear Optimization
Week 10	Data Visualization + Special Topics in Analytics

DELIVERABLES

This course has a consistent workload throughout the fast-paced quarter. In particular, be wary that the beginning of the course contains a lot of material as we establish the foundation and your comfort using Radiant.

It has a substantial amount of deliverables to ensure that students have enough practice with the material, both on fundamental aspects (via individual assignments) and more in-depth analyses (via group cases).

ASSIGNMENTS

There will be seven (7) weekly assignments in this course.

- Assignments are due via Canvas quizzes.
 - o All assignment Quizzes can be taken TWICE.
 - o This encourages the HW to be a learning tool.
 - o In your first attempt, you can try your answers without concern for your grade. Then, Canvas will show you which ones you got wrong. This allows you an opportunity to figure out why it was wrong or ask for clarification, and then re-submit the quiz.
- The Canvas quiz will automatically grade your answers, but you must still submit your all of your handwritten or Radiant work to the Canvas quiz. Proof of work can be shown using screenshots of Radiant, Word documents or PDFs, or Radiant HTML reports.
- The TA will check all student submissions for proof of their work.
- You are allowed to discuss homework topics at the conceptual level. If you collaborated with classmates on your homework, please note who you collaborated with at the top of your submitted homework.
- The quiz does not automatically provide partial credit. If you feel there should be partial credit for your homework, please email the TA within 3 DAYS of the homework due date.

GROUP CASES

There are 6 group cases due this course in Weeks 3, 4, 5, 6, 8, and 10.

- These group cases consist of a slightly more in-depth analysis in a specific business and industry setting. The goal of each case is for you to discuss the case in detail with your team, figure out which analyses are useful to apply, and then together agree upon recommendations. These recommendations should be backed up by solid arguments, evidence, and mathematical or data analysis as needed.
- The analysis will be slightly more difficult, but the grading of these cases will also focus on how you convey your analysis and findings. Communicating analyses is a very valuable aspect of business analytics. It takes practice to balance explaining analytics in-depth without overloading detail.
- Each case will require a ~2 page memo summarizing your analyses, along with an Appendix including the detailed analyses. The memo should be written in a

managerial style with limited use of jargon. Additional details and examples will be covered in Weeks 1 and 2.

READING // RADIANT TUTORIALS

From the textbook, there is required reading before each week (including before Week 1). Preparation reading for each week:

Date	Class Topic & Activities
Week 0	Radiant data exploration and visualization practice
Week 1	Section 1, Section 2.1, Kendall Crab Case (p. 56)
Week 2	Section 2.4 – 2.7, Section 3.1 – 3.4, San Carlos Case (p. 98)
Week 3	Section 3.7, Section 4
Week 4	Section 2.9, Section 6.1 – 6.4, review Hypothesis Testing Primer (on Canvas)
Week 5	Section 6.5 – 6.11, Oil Plus Case (p. 295), Exec Compensation (p. 297)
Week 6	Section 5 (incl. Gentle Lentil Case)
Week 7	
Week 8	Review Section 5 and Simulation Cheatsheet Handout (on Canvas)
Week 9	Section 7
Week 10	Section 9

Radiant Tutorials:

- On Canvas, at the top of the Modules page, is a list of all of the Radiant tutorials we will cover throughout this course. This is a very important resource to understand the main tool of the class, Radiant.
- Each week's Canvas overview will link to the few specific tutorials relevant for each lecture.

EXAMS

Midterm

- The midterm exam will be a take-home project report and a 3 minute presentation in Week 7.
- You will analyze one dataset thoroughly, using all of the key tools we have learned in Weeks 1-6. I will provide a set of existing datasets curated by me. Even better though, you can choose your own datasets that you are interested in analyzing from your work, past work, or industry domain of interest.

- You will be graded on the correctness of your application and interpretation of each topic's analysis. The specific midterm rubric will be made available by Week 5.
- The deliverables of this take-home data project will be a report summarizing the data and analyses, a few slides introducing your data and discussing 1-2 interesting analyses, and a brief 3 minute in-class presentation.
- The midterm project is a great takeaway from the class to illustrate on your resume or portfolio. It illustrates your ability to take an existing dataset and idea, and then complete all the necessary aspects of a thorough analysis to derive useful business insights.

Final

The final exam will occur during the week following the final class session and you will have 3 hours to complete the final exam. The final exam will be primarily a paper-based written exam, with short answer questions and problems to solve.

A few pages of written notes will be allowed (TBD), but the exam will not be designed to test your memorization of formulas. Rather, the final exam will test your ability to perform, design, and interpret data analysis. The final exam will ask you to interpret situations, design and describe your analysis approach, and use provided analytics output to determine insights or recommendations.

For your convenience, the final exam will include a cheat-sheet of potential formulas or calculations that could be needed; this cheat-sheet will be shared ahead of time as well.

CLASS PARTICIPATION

Your class participation grade is based on your responses to the required pre-lecture videos, and contribution to class discussions. To prepare for class, you should fully complete the required assignments and reading prior to each session. When class breaks into small groups for discussion, you should strive to be an active participant and contribute to the quality of the discussion. Please note that the frequency of your interventions in class is not a key criterion for effective class contribution.

AI + CHATGPT POLICY

ChatGPT and large-language AI models have become popular in the past couple years. They can answer simple statistical questions. Modern statistical analysts and programmers can use these, however, they must also be wary that they can make simple mistakes too.

If you find it helpful, you are welcome to try to use ChatGPT to offer advice or additional support on your homework, cases, and midterm projects. However, you must ultimately do the work on your own in Radiant and show the supporting evidence. And for the case writing, your group is expected to write the executive summary report and include analysis

on your own. The TA will be particularly vigilant in their grading of any non-sequitur output that a ChatGPT-type model might output.

Furthermore, for the midterm, you must write the report and answer the analytics questions on your own. Given that this is an MBA course, the goal is primarily your education and growth, not a particular letter grade. The midterm has been one of the best learning resources for students in the past, so I encourage you to write and analyze the midterm project primarily on your own. Furthermore, in the end, I do not think ChatGPT will offer too much towards completing the midterm.

Lastly, the final exam will be primarily paper-based, with no cell phone use and a limited use of a number of written notes. The final exam will emphasize your ability to interpret, understand, design, and utilize analyses, and less so running the analyses themselves.

COURSE SUPPORT

Business Analytics/Quant can be a difficult first-course at Rady, especially if you fall behind on a week. It can also be difficult in the first few weeks as you get comfortable with Radiant and the foundations of the class. We recognize this and have set up many avenues for support for this class.

TA Office Hours

- There will be TA office hours on multiple weekdays. These will be posted on the Canvas homepage.
- They will be on Zoom and have multiple “break-out rooms”. You can join with your study group, or other students there to allow them to be co-operative work sessions. In each breakout room, you can “raise your hand” on Zoom to request a TA to step in. The TA’s will be asked also to occasionally pop into your breakout rooms to see if they can help.

Canvas Discussions

- In the course Canvas, there is a Discussions section on the sidebar where students can ask questions and other students can help answer these questions to help and test your learning.
- Both I and a TA will work to provide answers quickly to clarify and answer the questions.

Professor Office Hours

- I will be available for 1:1 office hours every week on Wednesdays. See the Canvas homepage for details on how to schedule these. These will be on Zoom.

Extensions

- If difficult situations come up for you, please do not hesitate to email me and we can discuss ways to adjust deadlines.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <https://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.